

Linyi (Tiger) Hou

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EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign

Exp. May 2025
Aug 2017 - May 2021

SKILLS

Programming Languages	Python, C/C++, MATLAB/Simulink, Javascript
Libraries	numpy astropy scipy pybind11, Eigen OpenMP
Tools	Git, GitHub, GMAT, Slurm, ADQL
Operating Systems	Linux, MacOS, Windows, FreeRTOS

HIGHLIGHTS FOR GN&C POSITION

Deep Space Navigation

- ❖ Designed, implemented, and analyzed novel PNT algorithms culminating in three journal publications.
- ❖ Proficient in unscented Kalman filtering for astrodynamics applications.
- ❖ Experienced in the modeling and simulation of sensors (navigation cameras and star trackers), system dynamics (orbital mechanics), and the space environment (spherical harmonics).

UAV Navigation and Control

- ❖ Implemented an LQR controller and observer in FreeRTOS.
- ❖ Helped students formulate and implement state-space model controllers with various sensor configurations, including gyroscopes, laser rangefinders, optical flow sensors, and time-of-flight anchors.

Student Projects

- ❖ NASA RASC-AL Competition: led a team of 10+ peers to develop a reusable lunar lander concept.
- ❖ Space Grant Midwest High Power Rocketry Competition: led an avionics team to design, manufacture, and implement a flap-based roll control system that uses gyro and magnetometer data.

WORK EXPERIENCE

University of Illinois Urbana-Champaign

Teaching Assistant (AE 202, AE 443, AE 483)

Jun 2021 - present

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

Collins Aerospace

Software Engineering Intern

May 2019 - Aug 2019

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated builds in TortoiseSVN using Python, reducing need for manual integration.

RESEARCH

Astrodynamics and Planetary Exploration Group (APEX)

Advisor: Dr. Siegfried Eggli

Interstellar Navigation and Star Identification

Sep 2024 - present

- ❖ Developing star identification techniques for interstellar travel at relativistic speeds by combining traditional algorithms with clustering based on stellar classification.

Celestial Navigation to Supplement DSN Tracking

Jul 2024 - present

- ❖ Collaborating with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.
- ❖ Designing sensor tasking/scheduling algorithms with novel considerations for clustered observations and visibility constraints.
- ❖ Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars

Nov 2022 - present

- ❖ Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- ❖ Simulated star tracker observations and performed Monte Carlo analysis to assess the feasibility of reestablishing Earth pointing via the proposed method.

Aerospace Mission Analysis Laboratory

Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation (XNAV)

Jun 2021 - present

- ❖ Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- ❖ Created a photon time of arrival simulation tool in Python and validated results against the pulsar timing package [PINT](#) to within 50 μ s.
- ❖ Demonstrated the computational feasibility of XNAV in a lost in space scenario.

Velocity-based Initial Orbit Determination

Aug 2019 - Apr 2024

- ❖ Developed a novel initial orbit determination (IOD) technique using sequential range-rate measurements to distant stars.
- ❖ Assessed the performance of velocity IOD methods using Monte Carlo simulation.

HONORS AND AWARDS

Yee Fellowship

2021, 2024

Grainger College of Engineering, University of Illinois Urbana-Champaign

Aerospace Engineering Alumni Advisory Board Fellowship

2024

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46th AAS GN&C Conference

2024

American Astronautical Society (AAS) Rocky Mountain Section

David and Catherine Thompson Space Technology Scholarship

2020

American Institute of Aeronautics and Astronautics (AIAA)